
A Robot Policy Designed as a Weight-Space Artifact Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and causal intervention

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Abstract

1 Can model weights be made a first-class analysis artifact without sacrificing closed-
2 loop capability? We introduce χ -VLA, a 20.1M-parameter vision-language-action
3 policy whose learned map contains only bilinear, linear, and rational operations. Its
4 checkpoint therefore specifies an explicit rational tensor program. A mechanical
5 audit finds no softmax, standard activation, or ordinary normalization, and local
6 reconstructions agree with deployment near the float32 precision floor. We derive an
7 exact weight-based decomposition of each softmax-free attention layer, including
8 rational query-key normalization, and reduce its otherwise intractable coefficient
9 Gram to $O(d^2)$ storage. In the primary screen, all 36 trained block-head pairs
10 favor the rank-128 weight-derived subspace over fixed equal-rank random controls.
11 The full eight-block appendix screen finds the effect in 83/96 pairs and reveals
12 depth-specific negative controls. At the policy’s visual bond, a causally ranked
13 64-dimensional subspace preserves 18/20 closed-loop successes, while a random
14 subspace preserves none. The artifact is behaviorally nontrivial: the policy reaches
15 $93.7\% \pm 0.8\%$ on LIBERO-Object over three seeds. The first direct coefficient-edit
16 screen, however, fails its preregistered selectivity gate. We therefore present a
17 capable model artifact with exact layerwise structure and a causal analysis surface,
18 not yet a semantic or selectively editable weight-space interface.

19 1 Introduction

20 Neural checkpoints are usually treated as opaque containers for functions. Their weights can still
21 support post-hoc analysis, but softmax, ordinary activations, and input-dependent normalization
22 obstruct direct conversion of the complete learned map into an explicit polynomial or rational tensor
23 object. Bilinear language and vision models show that changing the primitives can expose useful
24 weight structure [3, 4]. Whether such a model artifact can remain behaviorally competitive in
25 continuous closed-loop control is unresolved.

26 We study two linked questions:

27 *Can the learned weights of a specialist VLA specify an explicit rational tensor*
28 *program, and can that artifact expose exact structure with a measurable causal*
29 *consequence?*

30 Our contributions are:

- 31 1. We construct and mechanically certify a complete VLA from tensor-compatible primitives.
32 A rational normalization supplies input-specific scaling without leaving the rational function
33 class.

2. We derive exact weight-based structure through each softmax-free attention layer and an $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
3. We test the artifact at behavioral scale. Exact subspaces beat fixed random controls, and a causally ranked visual subspace preserves 18/20 closed-loop successes while a random subspace preserves none.
4. We establish that the analyzed artifact is a capable policy, reaching $93.7\% \pm 0.8\%$ over three LIBERO-Object seeds, while reporting a language shortcut and a failed coefficient-surgery gate that bound the current interpretation.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The three matrices are a CP factorization of the order-three coefficient tensor.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

The deployed function is exactly rational. Projective coordinates carry each activation as a pair (N, D) representing N/D . Linear, bilinear, residual, and normalization operations update the numerator and denominator without introducing a non-rational learned operation.

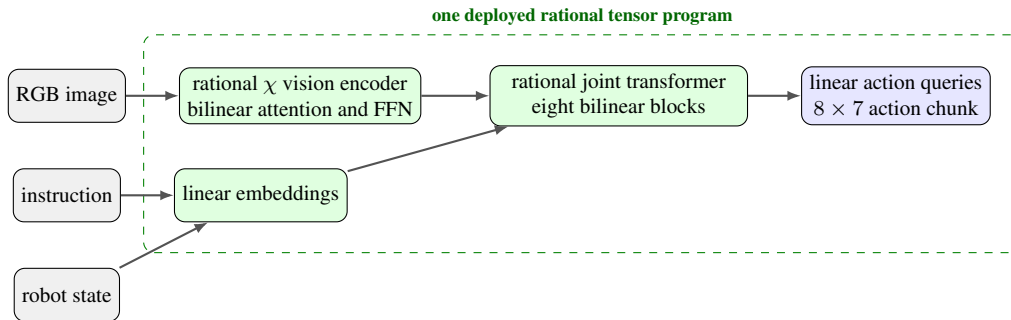


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

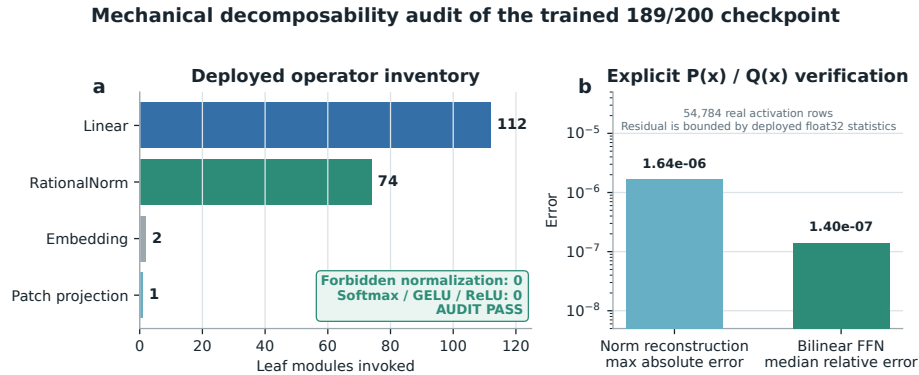


Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 The artifact is a capable closed-loop policy

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

85 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

86 The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%,
 87 and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed
 88 comparison.

89 Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3
 90 points above the mean of its members and 1.2 points above the strongest member. The gain is largest
 91 on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is
 92 not one 20.1M-parameter policy.

93 3.3 What this result does and does not establish

94 The result shows that the rational architecture supports high closed-loop specialist capability. It
 95 does not isolate the cost of decomposability. A decisive cost experiment must train a same-skeleton
 96 conventional transformer and staged partial-conversion controls using identical data, updates, seeds,
 97 and evaluation states. We treat that experiment as required before restoring the stronger title claim
 98 that convertibility is nearly free.

99 4 Exact weight-derived structure through attention

100 4.1 Exact layerwise construction

101 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 102 a high-order coefficient object determined entirely by the weights, mask, and fixed scaling. On a
 103 tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial
 104 agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

105 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 106 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 107 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 108 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 109 10^{-16} . Agreement with the real trained module ranges from 3.10×10^{-7} to 1.03×10^{-5} because
 110 the deployed implementation intentionally casts the statistic to float32.

111 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 112 values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. At toy multi-head scale, the
 113 reduced calculation matches brute-force materialization to 2.13×10^{-14} .

114 4.2 Trained policy result

115 We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final
 116 backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all
 117 evaluation examples. Real cached activations are used only to evaluate how faithfully the selected
 118 weight-derived subspaces preserve the layer’s computation.

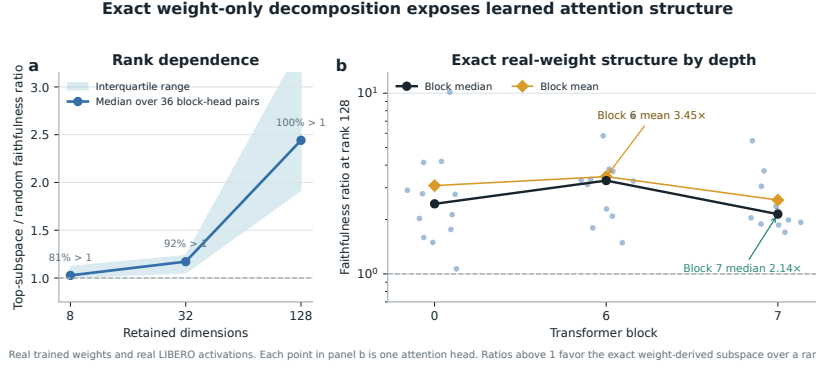


Figure 3: Exact weight-derived attention structure. Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. We compare the selected subspace with random equal-width subspaces in offline action reconstruction and closed-loop intervention. This ranking uses held-out activations and downstream sensitivity. It is causal, but not a weight-only discovery result.

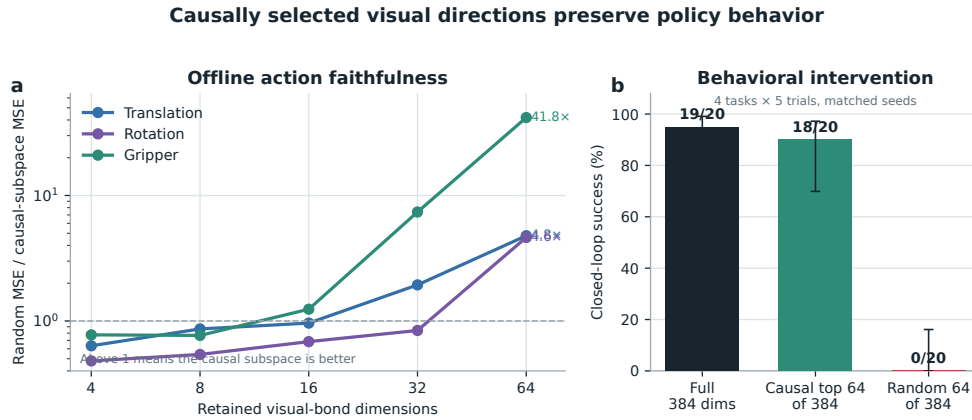


Figure 4: Causally selected visual directions preserve policy behavior. The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

134 With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error relative
135 to random by $4.8\times$ for translation, $4.6\times$ for rotation, and $41.8\times$ for the gripper. In closed loop, the
136 full representation succeeds in 19/20 trials, the selected subspace in 18/20, and a random subspace
137 in 0/20.

138 This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and
139 twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream
140 sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same
141 full-suite closed-loop intervention.

142 5.2 Counterfactual instruction test

143 High success does not guarantee robust language use. We hold the image and robot state fixed
144 and replace the instruction with one naming another visible object. Across 80 canonical one-step
145 trials, the predicted reach direction shifts toward the newly named object in only 15.0%. Mean
146 directional cosine to the named object falls from 0.525 under the true instruction to 0.299 under the
147 counterfactual instruction, while cosine to the original target remains 0.464. In an earlier 40-step
148 closed-loop screen, 33.3% finish closer to the new object and 55.6% move measurably toward it.

149 The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target,
150 so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations
151 improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-
152 loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy
153 range.

154 This negative result narrows the VLA claim. The current model is a capable language-conditioned
155 specialist, but it is not yet a robust instruction-grounded policy. LIBERO-CF independently documents
156 the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the
157 shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively
158 repair that shortcut.

159 6 Related work and limitations

160 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
161 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
162 weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained
163 VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a
164 policy designed as a rational tensor program. Tensor and polynomial networks provide the broader
165 multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success
166 can coexist with weak language use [9–11].

167 **Limitations and conclusion.** No same-skeleton conventional twin yet measures conversion cost.
168 The benchmark is one specialist suite, the policy fails a counterfactual instruction test, and exact
169 decomposition is layerwise rather than end to end. The closed-loop subspace test uses one checkpoint
170 and twenty trials, while the first direct coefficient edit fails its selectivity gate. The defensible result
171 is therefore narrow: rational restrictions do not prevent competitive LIBERO-Object control, and
172 they provide a precise weight-derived analysis surface. Broader grounding, matched controls, and
173 validated coefficient edits remain necessary.

174 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
175 certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are
176 unavailable.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

208 A.2 Per-task ensemble behavior

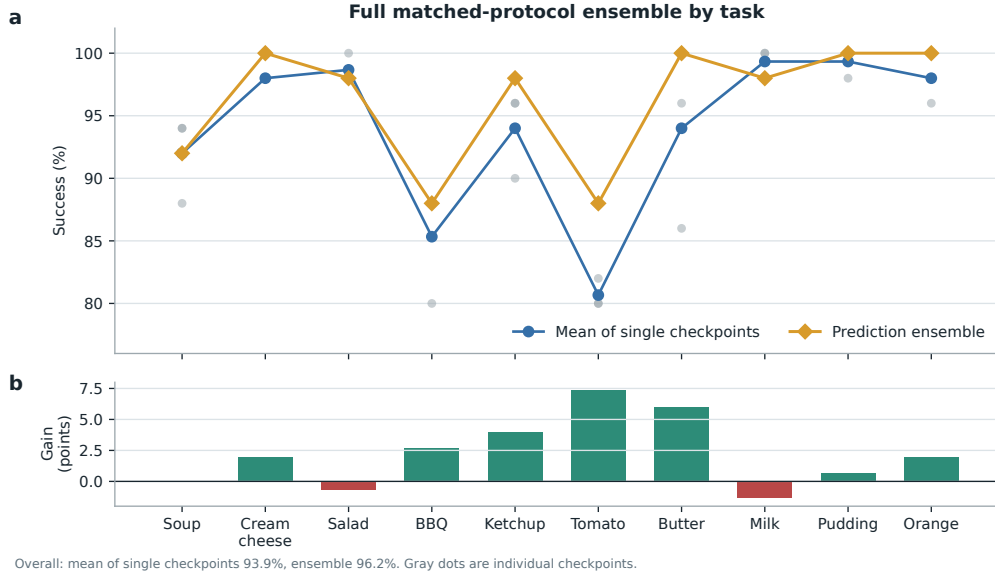


Figure 5: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

209 A.3 Product-routing multimodal head

210 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

211 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 212 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$
 213 and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head.
 214 Product routing remains a construction result until it wins a controlled genuinely multimodal task.

215 A.4 Development interventions

216 Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven
 217 corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces
 218 overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points.
 219 A reward audit shows that accumulated step cost makes every successful trajectory have negative
 220 return. After removing that term and repairing the terminal boundary, one pilot improves by two
 221 points. These are development diagnostics, not validated paper contributions.

What the July 30 intervention experiments taught us

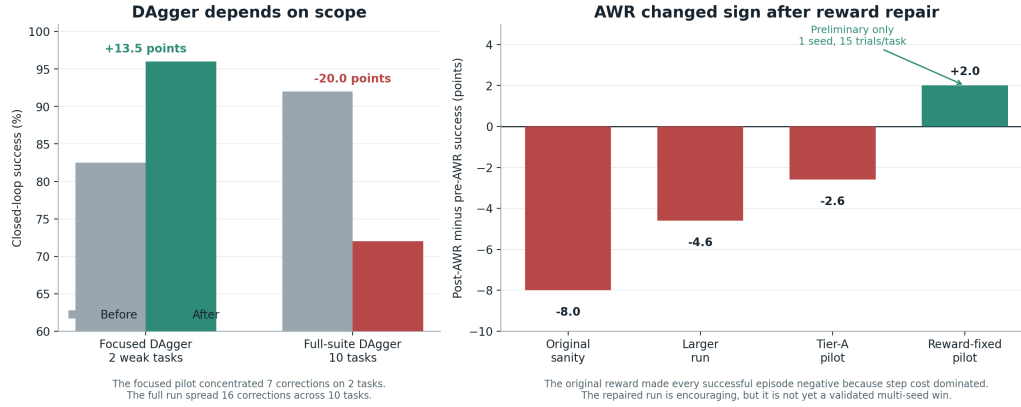


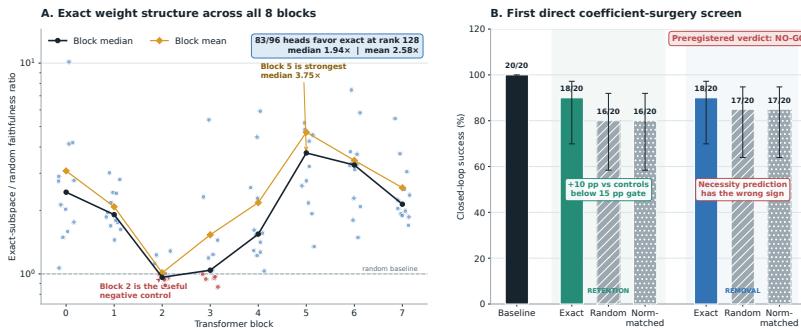
Figure 6: **Development interventions reveal scope and reward sensitivity.** Focused DAGger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

222 A.5 Expanded attention screen and first direct edit

223 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 224 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 225 is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96
 226 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at
 227 the wider tested rank.

228 The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-
 229 subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This
 230 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20,
 231 compared with 17/20 for both random controls, which is the wrong direction for the necessity
 232 prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and
 233 value column edits.

Weights expose reproducible structure, but the first direct edit is not yet selective



A: 96 block-head pairs, fixed random-control banks, rank 128. Exact directions come from weights; cached activations only score layer faithfulness. B: 4 tasks × 5 paired canonical episodes. Direct QKV weight edits, no hooks, gradients, retraining, or rollout fitting. Error bars are 95% Wilson intervals.

Figure 7: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

234 **A.6 Numerical scope of rational conversion**

235 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
236 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
237 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
238 establish practical numerical safety under distribution shift.